FINAL PROJECT

ENGR 2 · GUS · FALL 2026

Transamerica Pyramid Lamp

An 18-inch PLA and plywood scale model of San Francisco’s pyramid that lights up from the inside.

- Prusa 3D printer
- Epilog laser cutter
- Carvey CNC mill



3D model: “Transamerica Pyramid” by seppi, SketchUp 3D Warehouse.

01 · THE IDEA

A scale model that’s also a lamp

- **18 in tall** — about 1:570 scale of the 853 ft tower.
- **Glows from inside:** an LED light shines through laser-cut windows backed with frosted acrylic.
- **Stands on its own block:** a printed truss foundation set into a carved street-corner base.

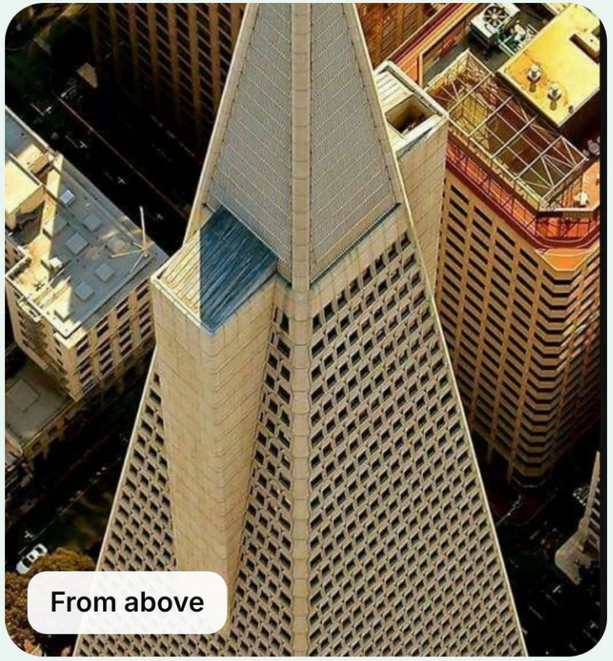
WHY THIS ONE?

I just like it. It’s a beautiful building, and the engineering is cool — that taper, and the truss base it all stands on.

48 floors | 3,678 windows | Finished 1972



Going up



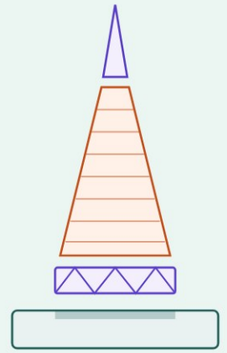
From above



Bummer & Lazarus, San Francisco’s famous 1860s stray dogs, have a plaque in Redwood Park at the pyramid’s foot.

02 · TOOLS & BREAKDOWN

Three machines, one building



PRUSA 3D PRINTER

Truss foundation, spire cap and window detailing.

Material	PLA, about 250 g
Machine time	~12 h, run overnight

EPILOG LASER CUTTER

Four tapered plywood walls with window cutouts, plus frosted acrylic windows behind them.

Material	⅛″ Baltic birch, ⅛″ frosted acrylic
Machine time	~2 h

CARVEY CNC MILL

8×8 in site base: street, sidewalk and a slot the model sits in.

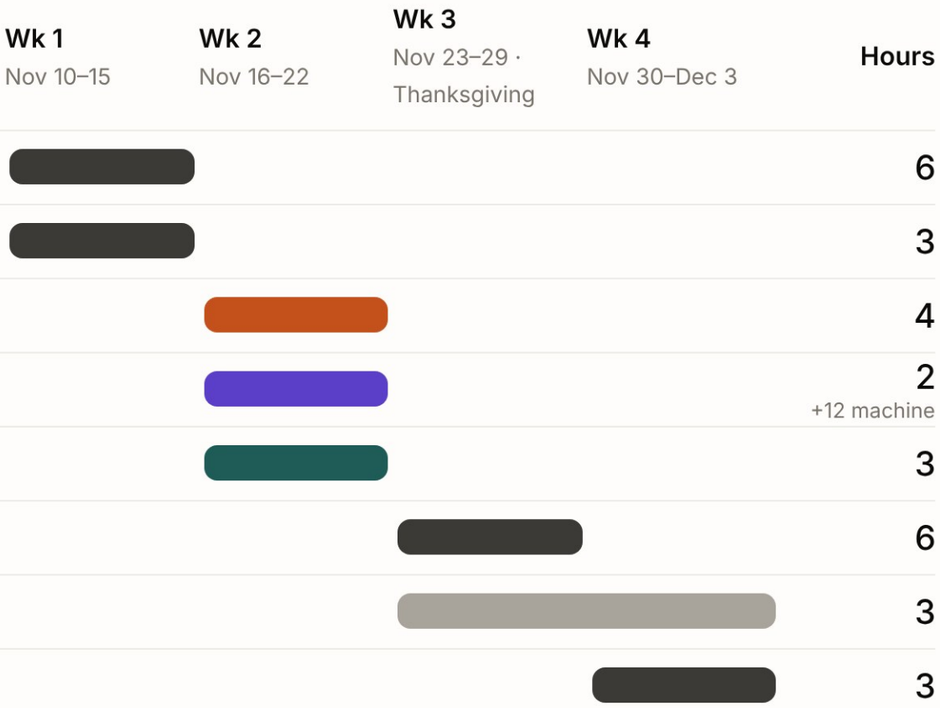
Material	¾″ Baltic birch
Machine time	~1.5 h

How it fits: the printed foundation keys into the carved slot, the walls glue onto the foundation, acrylic sits behind the window cutouts, and an LED puck light sits in the hollow core.

Extra tools: none — just glue, clamps and sandpaper.

03 · TIME PLAN & BUDGET

Nov 10 → Dec 3, about 30 hours



Machine work happens in class sessions before Thanksgiving week; about 6–8 h a week outside class; long prints run overnight.

EXPECTED BUDGET (RETAIL)

⅛″ Baltic birch, 12×24 3 sheets × $7.94	$23.82
⅛″ frosted cast acrylic, 12×24 1 sheet	$24.79
Prusament PLA ~250 g of a $28.79 kg spool	$7.20
¾″ Baltic birch base, 8×8 in share of a $52.99 sheet	$3.77
USB LED puck lights 3-pack	$10.98
Wood glue + CA glue estimate	$8.00
Total	**$78.56**